

LSTM vs RNN

F1 & Accuracy:

RNN:

Accuracy : 0.5343137254901961
F1 Score : 0.5159032024363004

LSTM:

Accuracy : 0.5318627450980392
F1 Score : 0.5205250806572143

Classification report :

RNN:

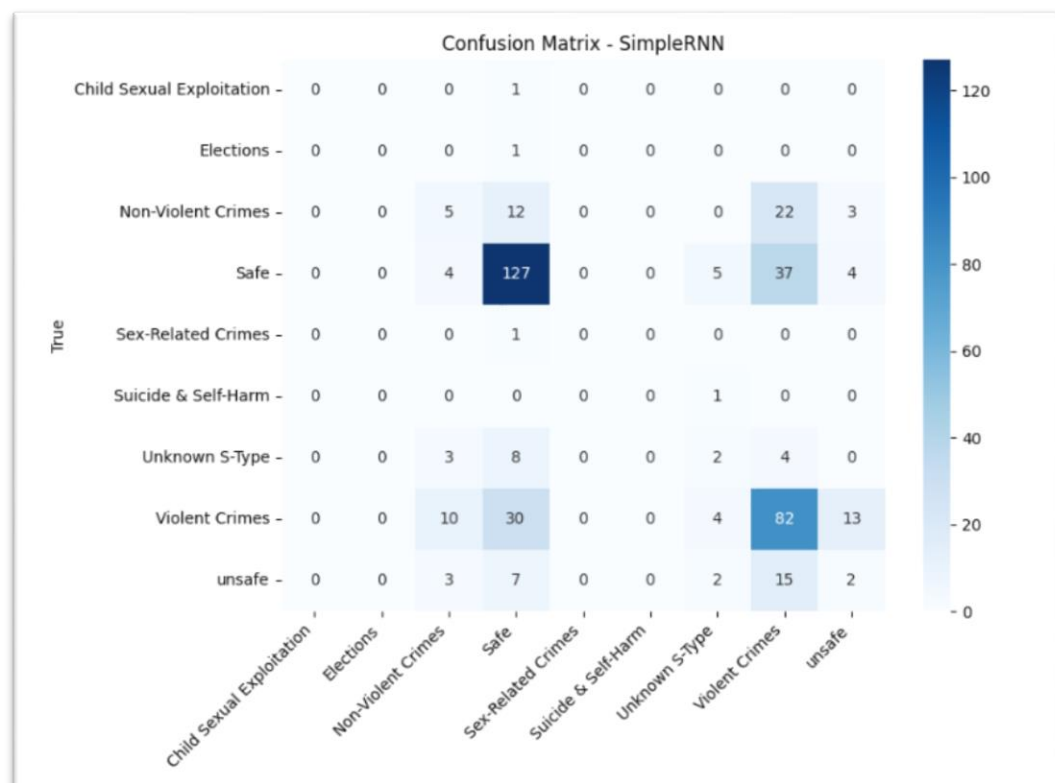
	precision	recall	f1-score	support
Child Sexual Exploitation	0.00	0.00	0.00	1
Elections	0.00	0.00	0.00	1
Non-Violent Crimes	0.20	0.12	0.15	42
Safe	0.68	0.72	0.70	177
Sex-Related Crimes	0.00	0.00	0.00	1
Suicide & Self-Harm	0.00	0.00	0.00	1
Unknown S-Type	0.14	0.12	0.13	17
Violent Crimes	0.51	0.59	0.55	139
unsafe	0.09	0.07	0.08	29
accuracy			0.53	408
macro avg	0.18	0.18	0.18	408
weighted avg	0.50	0.53	0.52	408

LSTM:

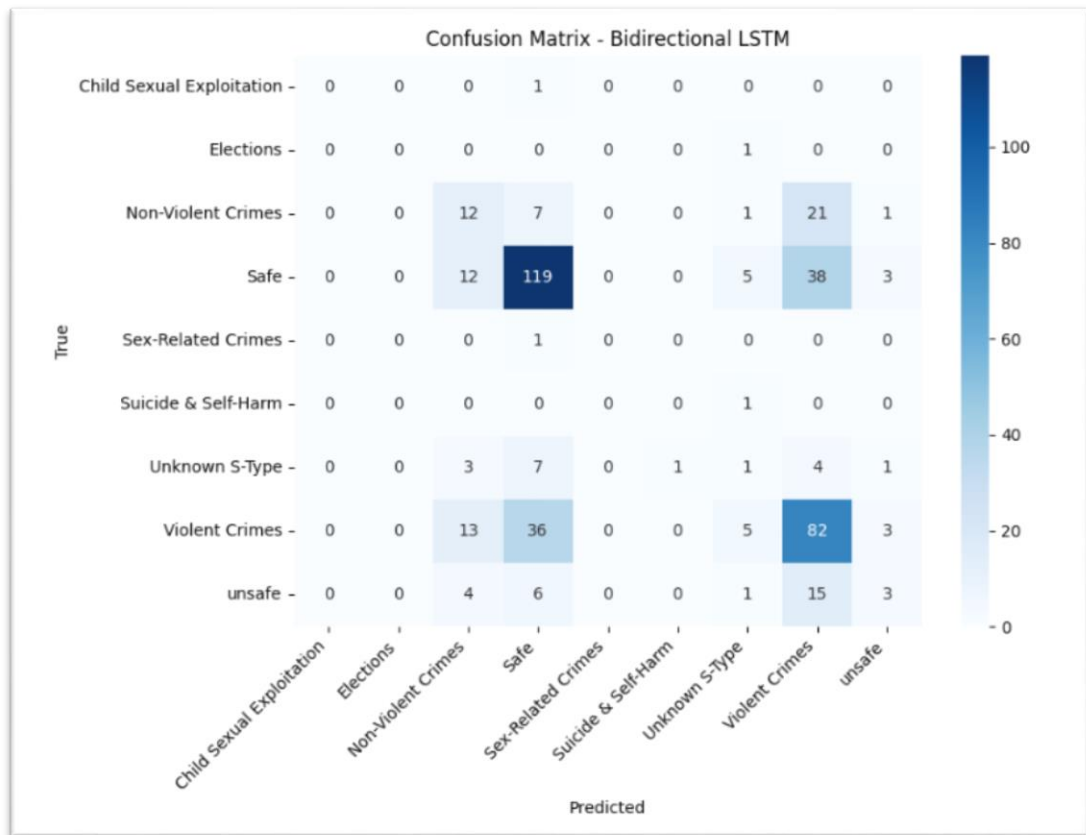
	precision	recall	f1-score	support
Child Sexual Exploitation	0.00	0.00	0.00	1
Elections	0.00	0.00	0.00	1
Non-Violent Crimes	0.27	0.29	0.28	42
Safe	0.67	0.67	0.67	177
Sex-Related Crimes	0.00	0.00	0.00	1
Suicide & Self-Harm	0.00	0.00	0.00	1
Unknown S-Type	0.07	0.06	0.06	17
Violent Crimes	0.51	0.59	0.55	139
unsafe	0.27	0.10	0.15	29
accuracy			0.53	408
macro avg	0.20	0.19	0.19	408
weighted avg	0.52	0.53	0.52	408

Confusion matrix:

RNN:

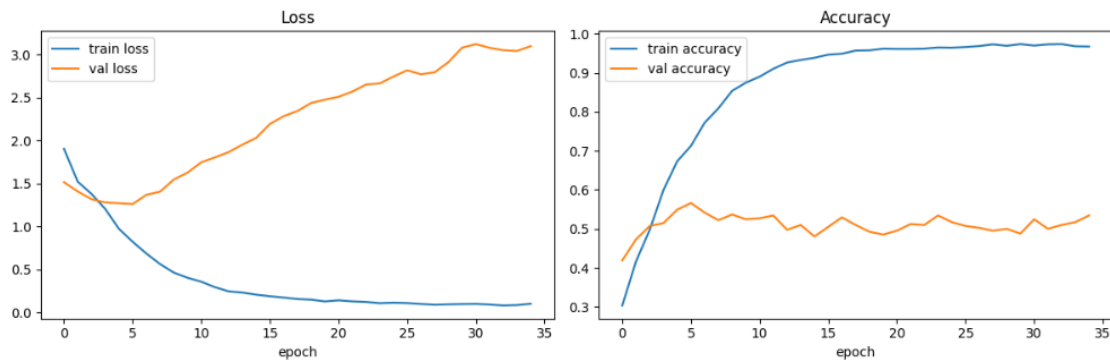


LSTM:

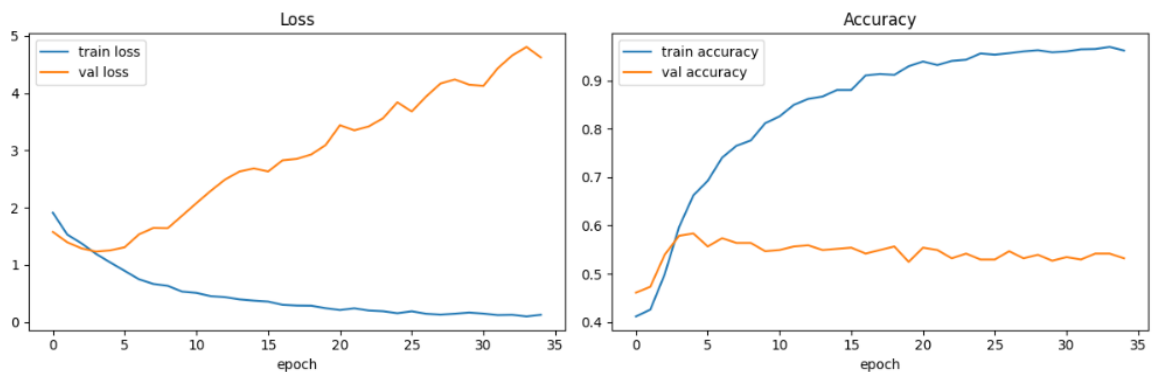


Loss Graphs:

RNN



LSTM



Conclusion:

Lstm is little more accurate because it offers more control on memory and amount saved information but overall the accuracy is not good bec the data is imbalanced and still needs more prerocessing